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**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks: 30

Annexure A

Constraint-Based Problem Solving

Declaration of Conduct:

(192372301)

Gr. Charan Teja (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Gr. Teja

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Nurse Assignment using Back Tracking Search

Problem statement:

A hospital must assign 3 nurses (N_1, N_2, N_3) to 3 wards (ICU, general, Emergency) while satisfying the given constraints

Constraints

1. N_1 cannot be assigned to emergency
2. N_2 must be assigned before N_3 according to ward order $ICU < general < emergency$
3. Only one nurse can be assigned to one ward
4. N_3 cannot be assigned to ICU.
5. Every nurse must be assigned exactly one ward

Back-tracking Search:

Back tracking is a search algorithm that assigns value one by one - if a constraint is violated, it returns (back-tracks) to previous assignment and tries another possibility.

Step by Step Search

Step 1: Assign N_1

Possible wards:

- ICU
- general
- Emergency

choose:

$N_1 \rightarrow \text{ICU}$

Current Assignment:

Nurse	ward
N_1	ICU

Step 2: Assign N_2

remaining wards

- general
- Emergency

Try:

$N_2 \rightarrow \text{general}$

Current Assignment:

Nurse	ward
N_1	ICU
N_2	General

constraint checks:

- Only one nurse per ward
- N_2 assigned before N_3

Step 3: Assign N_3

remaining ward:

- Emergency

check constraints

- N_3 cannot be ICU.
- ward available.

Autonomous Rescue Drone using AI Search

Problem Statement:

An autonomous rescue drone must locate a person inside a disaster-affected building represented as a grid. Some rooms are blocked, some are visible, and the drone can only observe nearby cells.

(a) Constraint Analysis

1- movement Constraint:-

The drone can move:

- up
- down
- left
- right

effect:- The search is limited to four possibilities at each step.

2- Blocked cells:-

Some rooms contain obstacles

effect:- The drone cannot enter blocked cells and must search for an alternative path.

3- limited sensing

The drone only observes adjacent cells.

effect:-

The drone does not know the complete map initially and updates its map while exploring.

Assign:

$N_3 \rightarrow \text{emergency}$

Constraint Verification

constraint	status
N_1 not in emergency	satisfied
N_2 before N_3	general before emergency
One nurse per ward	✓
N_3 not in ICU	✓
every nurse assigned one ward	✓

Search Tree

Start

$N_1 \rightarrow \text{ICU}$

$N_2 \rightarrow \text{general}$

$N_3 \rightarrow \text{Emergency}$

Valid solution

Final Schedule

Nurse	Assigned ward
N_1	ICU
N_2	General
N_3	Emergency

Conclusion:

Backtracking successfully assigns all nurses while satisfying every constraint whenever a constraint is violated, algorithm backtracks and tries another assignment until a valid solution is found.